
UNIT 1: DATA WRANGLING AND PROFILING

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1.0 INTRODUCTION

This is the first unit of the course on Data Wrangling and Visualisation. It introduces the basic details of Data wrangling and profiling of the data. Data wrangling is defined as a process, which converts the raw data into more reliable, usable data containing a lesser number of errors and missing values. Thus, the data, after being processed by various methods of data wrangling, can be used for analysis readily. Further, you would like to extract information about data to know the data types of various variables, their basic distribution, the relationships between the sets of data, etc. This is known as data profiling. The unit provides detailed discussion of the data wrangling and profiling. Data semantic process tries to find the context in which data is being generated. This unit provides detailed information about data semantics. The unit also introduces some of the basic tools for data wrangling. Finally, the unit discusses summary statistics

1.1 OBJECTIVES

At the end of this unit, you should be able to:

- define data wrangling and its need;
- discuss data modelling concepts;
- explain data semantics;
- define the tools of data wrangling
- perform some basic operations related to set-based profiling;
- explain the use of data integration.

1.2 BASIC CONCEPTS

Data wrangling and profiling are the first steps of data analysis process. It is performed to enhance data quality and accuracy, resulting in better analysis. Let us first define the two terms used in this unit - Data Wrangling and Profiling in the next sub-section.

1.2.1 Definition

Data wrangling is the process in which raw data is converted to a more usable form, which can be used for analysis readily. Data Wrangling is also known as Data munging.

Data Profiling: Data types, formats, value distributions, patterns, and their interactions, etc., are better understood through this approach.

Raw data often has lot of errors, is not accessible for every row and column of the data (missing values), and sometimes is not in an easily usable format. Prior to analysis, it cleans, shapes, and improves the input raw data. This procedure requires cleaning the data by eliminating or fixing any errors, duplications, and inconsistencies. Data wrangling also helps in converting the raw data into tabular form.

Through data wrangling, this raw data is converted into a format suitable for analysis. Hence the process of data wrangling enhances the overall quality of the data and makes it possible to examine more accurate and trustworthy data. This improves analysis while also saving users time and effort.

Furthermore, the accuracy and validity of the results or models of analysis are guaranteed if data manipulation is employed to improve the data, such as by filling in the missing value. By making raw data more accessible and pertinent, data wrangling enables analysts and data scientists to get crucial insights more rapidly and accurately.

1.2.2 Need of Data Wrangling

The relevance of data wrangling continues to grow for several reasons:

1. **Volume, type and variety of Data:** Data generation volume has increased multi fold by the evolution of social media, the internet, IoT devices, and many other sources must be appropriately managed and analysed by organisations. Data wrangling enables efficient handling of this exponentially massive and heterogeneous data.
2. **Advanced Analytics and Artificial Intelligence:** The advancements in analytics and artificial intelligence (AI) require high-quality data. Data wrangling ensures that the data is clean, accurate, and structured data used in these complex models so that AI and machine learning programs can work successfully.
3. **Faster, informed and effective Decision Making:** In today's fast-paced world, staying ahead of the competition requires making prompt, well-informed decisions. Administrations can analyse data more efficiently and gain understandings which is not possible without data wrangling, which expedites data preparation.
4. **Administration of data and adherence to different Data Protection Rules and regulations:** Organizations need to ensure that data is properly preserved

and processed in the light of the growing number of data protection and usage regulations, such as the Rules for Digital Personal Data Protection, 2025.

5. Data wrangling ensures adherence by arranging and cleansing data in line with these rules and regulations.
6. **Data Excellence and Accurateness:** The reliability of data analytics is significantly influenced by the quality and correctness of the underlying data. Data wrangling enhances the quality and correctness of the data, strengthening the validity of the conclusions that are drawn from it.

1.3 PROCESS OF DATA WRANGLING

Data wrangling involves several steps to convert raw data into a form which can be readily used. Following steps show how data wrangling works:

1. Collection

First step in data wrangling is collecting required data from several sources. These sources include files, databases, IoT device data input, data extraction from websites, and numerous other data streams. Collected data may be semi-structured (like JSON/XML files), unstructured (like text documents, images, audio or video files), or structured (such as SQL databases).

2. Cleaning

Cleaning process starts as soon as the data is collected. Oversights, inconsistencies, and duplication are removed in this step since they may skew analysis results. It might include:

- eliminating information that is not important to the analysis.
- fixing data mistakes, like misspellings, inaccurate and Null values.
- lower case and upper case normalisation,
- addressing missing values by eliminating them, assigning them to other data points, or using statistical techniques to estimate them.
- identifying and fixing inconsistencies, such as different date or currency formats.

3. Structuring

After cleaning, data must be organised in a way that makes analysis easier. Typically means converting raw data (unstructured or semi-structured) into a format which could be directly used for analysis such as CSV files or database tables. It includes

- converting data into structured fields.
- ensuring that data is in consistent formats and units by normalising it.
- information transformation, such as converting text to lowercase, to prepare it for analysis.

4. Enriching

In order to improve dataset's analytical usefulness, data enrichment process adds new information or context. It can include:

- merging data from different sources to produce a dataset that is more comprehensive.
- adding new features which could provide more information when analysis is done.

However, this step must ensure that the analysis findings are consistent.

5. Validating

Validation makes sure that data is correct and quality of the data once data is cleaned, structured, and enriched. This involves:

- ensures that a database's primary and foreign keys match, among other data integrity tests.
- data compliance with defined standards and norms is ensured by quality assurance testing.

6. Storing

After preparation and organisation, data is efficiently arranged and securely stored for reporting, querying and analysis in a data inventory like database or data warehouse

7. Documentation

Documentation is essential at every stage of the data processing process. It tracks the process, modifications, and choices made in context of the data. This helps in understanding the data analysis method, auditing, and reproducibility

1.4 BENEFITS OF DATA WRANGLING

Data wrangling has numerous advantages that significantly increase the worth of data for companies and organisations. Data wrangling opens the door to more accurate, effective, and insightful research by transforming unstructured data into more structured and clean format. Here are a few specific advantages of data wrangling:

1) Improved Data Quality

The notable improvement of data quality is one of the primary benefits of data wrangling. Inaccuracies, inconsistencies, missing values, and repetitions are common in raw data, which can skew analysis and can lead to inaccurate conclusions. To solve these problems, the data wrangling process - cleaning is used, and later the validation step assures that the accurate, consistent and reliable data is used in the analysis. To obtain reliable insights so that decision makers can make wise judgments, high-quality data is essential.

2) Enhanced Analytical Efficiency

By streamlining the data preparation procedure, data wrangling improves the effectiveness of data analysis. By automating repetitive operations and utilising advanced data cleansing and organising tools, data scientists and analysts spend more time on analytical work and less time on data preparation. This increases productivity, which speeds up the analytical process and enables analysts to examine more data at once and finish more complex projects in less time.

3) Facilitation of Advanced Analytics and Machine Learning

Clean, structured and organised data is necessary for use in complex analytics and machine learning models to produce accurate results. Data wrangling transforms

raw data in a format that these types of models can easily process, enabling more sophisticated insights. For these complex applications, such as trend analysis, customer segmentation, and standard predictive analytics, data wrangling ensures that the underlying data is in the most effective possible form so that it can produce more accurate and useful results.

4) **Data Integration from Multiple Sources**

Data in the modern digital age originates from various kinds of sources which are diverse, such as social networking sites, enterprise systems, Internet of Things devices, and more. Data wrangling assists in combining information from various sources, filtering, standardising formats, and resolving discrepancies to produce a coherent dataset to increase its efficacy and facilitate integration. This integration is essential for in-depth analysis since it considers all relevant facts and provides more comprehensive perspective and helps in effective decision making.

5) **Compliance and Data Governance**

Since data privacy, protection, and regulatory compliance are becoming increasingly important, data wrangling is essential to ensure that data is handled and processed in accordance with ethical and legal requirements. Organisations can better manage their data governance obligations, ensuring that sensitive and personal data is managed safely, correctly and lowering the risk of compliance concerns, by organising and cleansing their data.

6) **Empowered Decision-Making**

The fundamental objective of data wrangling is to support better and effective decision-making. Data wrangling enables organisations to make decisions based on precise and comprehensive information through proper data cleaning, structure, and enrichment. This results in improved organisational policies, increased operational performance, and a stronger competitive advantage in the market.

7) **Scalability**

Data wrangling procedures and technologies can be scaled to match the increasing demands of enterprises, which must manage a significantly larger volume of data as they expand. Good data wrangling techniques ensure that companies can handle bigger datasets without seeing a corresponding rise in errors or deterioration in the quality of insights.

After discussing about the basics of data wrangling, its process and benefits, in the next section, let us discuss a very important aspect of data, which is the meaning of each data element.

Check Your Progress 1

1. Why is data wrangling important for AI and Machine Learning?

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2. What is the primary goal of data wrangling? Define data profiling.

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3. Why is data wrangling needed?

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1.5 DATA SEMANTICS

Data Semantics represents the meaning and structure of data. By transforming unstructured, confusing data into information with a common meaning, it helps you enforce data integrity and data integration and enhances data analysis.

Semantic data models can be produced via an abstraction process that selects real-world data elements and creates links between these attributes to generate organised, meaningful data.

Sometimes, raw data can be confusing. Additionally, metadata—data about data—is frequently poorly defined. Sometimes a higher-level abstraction of the data is required in order to represent the data in an organised manner. This leads to data format or standardisation in addition to making the data's meaning more understandable. Better data interpretation and integration result from this. The notion of data standardisation will be further explained by the example that follows:

Example: Numerous department stores in several cities are part of a nationwide chain. It keeps track of retail purchases made by several clients in a single data file. The raw sample data of these stores is given in the following Table:

Table 1: A Sample Dataset of Store data

ID	Name	Gender	Date of Purchase	Category	Purchase Amount	City
101	Rajiv	M	13/03/26	Laptop	15000	Delhi
102	Nita	Female	2026-01-13	Computer	19000	New Delhi
103	Madan	male	19-Jan-2026	Electrical	21000	DEL
104	Nazia	F	25/11/2025	Elect	25000	Madras
105	Robert	Male	2025/04/19	Comp	37000	Chennai

You may observe the following problems in this data:

1. The Gender attribute is represented using abbreviations, and the first letter is sometimes in lowercase.
2. Date of Purchase is in several inconsistent formats.
3. Data in the Category and City attributes are not consistent.

How can semantic abstraction be used to convert this data to consistent data? The following table suggests these semantic abstractions:

Table 2: Data Standardisation for the data of Table 1

Attribute	Raw Data	Data Standardised to
Gender	M or male or Male	MALE
	Female or F	FEMALE
Date_of_Purchase	All date types	Format dd-mm-yyyy
Category	Laptop or Computer or Comp	Computer
	Electrical or Elect	Electrical
City	Delhi, New Delhi, DEL	DELHI
	Madras, Chennai	Chennai

The following program uses the pandas library to perform these semantic abstractions:

```
#Program 1 Semantic Abstraction (as shown in Table 2) of the
data of Table 1
import pandas as pd
# Raw dataset of Store
store_data = {
    "ID": ["101", "102", "103", "104", "105"],
    "Name": ["Rajiv", "Nita", "Madan", "Nazia", "Robert"],
    "Gender": ["M", "Female", "male", "F", "Male"],
    "Date_of_Purchase": ["13/03/26", "2026-01-13", "19-Jan-
2026", "25/11/2025", "2025/04/19"],
    "Category":
    ["Laptop", "Computer", "Electrical", "Elect", "Comp"],
    "Purchase_Amount": [15000, 19000, 21000, 25000, 37000],
    "City": ["Delhi", "NewDelhi", "DEL", "Madras", "Chennai"]
}
#Create the dataframe with the name storedf from the
store_data
storedf = pd.DataFrame(store_data)
# Perform the Gender attribute mapping
gender_map = {"M": "MALE", "male": "MALE", "Male": "MALE", "F":
"FEMALE", "Female": "FEMALE"}
}
storedf["Gender"] = storedf["Gender"].map(gender_map)
# Convert Date_of_Purchase into datetime data type
storedf["Date_of_Purchase"] =
pd.to_datetime(storedf["Date_of_Purchase"], dayfirst=True)
# Convert datetime datatype format into dd-mm-yyyy format
storedf["Date_of_Purchase"] =
storedf["Date_of_Purchase"].dt.strftime("%d-%m-%Y")
# Perform the Category mapping
category_map = {"Laptop": "Computer", "Computer":
"Computer", "Comp": "Computer", "Elect":
"Electrical", "Electrical": "Electrical"}
}
storedf["Category"] = storedf["Category"].map(category_map)
# Perform the City mapping
city_map = {"Delhi": "Delhi", "New Delhi": "Delhi", "DEL":
"Delhi", "Madras": "Chennai", "Chennai": "Chennai"}
```

```
}  
storedf["City"] = storedf["City"].map(city_map)  
#print the dataset after Semantic Abstraction  
print("Store DataSet after Semantic Abstraction")  
print(storedf)
```

Program 1: Semantic Abstraction

At the first statement, the *pandas* library is imported as *pd* so that you can use its classes and methods. Next, a data set called *store_data* is constructed that contains the information displayed in Table 1. The dataframe object named *storedf* is then created by calling the dataframe constructor function (*pd.DataFrame(store_data)*).

The attribute mappings are then completed, first by creating a mapping and then by applying it to the created dataframe. These mappings are made for the properties of gender, category, and city. The data is first transformed to datetime format for the *Date_of_Purchase* attribute, and then it is converted to *dd-mm-yyyy* format.

You will receive the following abstracted dataset for the store data after running program 1.

Dataset after Semantic Abstraction

ID	Name	Gender	Date of Purchase	Category	Purchase Amount	City
101	Rajiv	MALE	13-03-2026	Computer	15000	Delhi
102	Nita	FEMALE	13-01-2026	Computer	19000	Delhi
103	Madan	MALE	19-01-2026	Electrical	21000	Delhi
104	Nazia	FEMALE	25-11-2025	Electrical	25000	Chennai
105	Robert	MALE	19-04-2025	Computer	37000	Chennai

Thus, the data standardisation can help in producing better quality data, as it removes ambiguity in data to produce more consistent data that can be used for better data analysis.

1.5.1 Data Semantics Process

The data semantic process is the generic process that operates on raw data to make it more meaningful and interpretable. The following example shows the basic steps that may be used in data semantic process

Example:

1. **Data Collection:** Raw data is gathered from sources. For example, the following raw data may be obtained as a series of data bytes:

Name | Value
A | 25
B | 30

2. **Data Structuring:** The raw data is organised into a structured format, such as tables, JSON, etc. The first record of the raw data, as shown in point 1 above, is organised as JSON object as:

```
{  
  "person": "A",  
  "value": 25  
}
```

It is still unclear what “person” and “value” means.

3. **Assigning Meaning(Semantic Annotation):** we will define what each data field means. For example:

```
{
  "person_name": "A",
  "age": 25
}
```

Now “25” clearly means it is the age of the person whose name is given.

4. **Using Ontology/Schema:** we define relationships and rules like (Age-> must be a number,
Age-> belongs to a person;
Person→has→Age

Data Interpretation: Now system can understand and use the data meaningfully. For example, you can interpret a person as an adult or a minor as:

if Age of a person >18, “he/she is adult” else “minor”

1.5.2 Semantic Wrangling using Data Abstraction

Sometimes the available data may need to be generalised or aggregated into different categories. In such situations, you may use data abstraction terms like “is a” (generalisation), “has a” (aggregation), and “instance of” (classification) to perform semantic wrangling. The following semantic relationships describe the connections between the attributes:

- **Generalisation (Is A):** For example, a manager “is a” type of employee. You can establish generalisation (“is a”) for a more general object.
- **Aggregation (Has A):** Combine multiple component objects or attributes to define a new object. For example, an employee “has” a name, age, and contact details.
- **Classification (Instance Of):** Sort certain items into groups or “instances of” according to common traits.

Modeling Semantic Data: You can create a semantic data model by utilising the stated abstraction approaches to specify the context and meaning of the data. In addition, you can add *metadata*, which ensures a common understanding and use. Please note that metadata can also explain the relationships, significance, and limitations of the data. Finally, you can create a *common terminology*, which ensures uniform interpretation across systems and stakeholders.

Application and Integration of Data: Data Integration provides a uniform and consistent perspective across many data sources by constructing a semantic layer that serves as a business representation of the data.

The following example shows the use of abstraction in Python.

Example: Consider the following data of 6 customers of a departmental store:

ConsumerName	Model	ProductType
Manish	Reno	Smartphone

ConsumerName	Model	ProductType
Meenu	Thinkpad	Laptop
Rafiq	Galaxy	Tablet
Samuel	MacBookAir	Laptop
Vikrant	Edge60	Smartphone
Kiara	Pad3	Tablet

The company want to analyse the data on the basis of generalised categories of ProductType data as follows:

SimpleMobile, Smartphone	Mobiles
Laptop, Tablet	Portable Computing

Write a Python program that performs the required abstraction and also creates a generalised new category in the data.

Solution: Program 2 does the required tasks.

```
import pandas as pd
# Dataset contains List of Dictionaries
ConsumerElectronics = [
    {"ConsumerName": "Manish", "Model": "Reno",
     "ProductType": "Smartphone"},
    {"ConsumerName": "Meenu", "Model": "Thinkpad",
     "ProductType": "Laptop"},
    {"ConsumerName": "Rafiq", "Model": "Galaxy",
     "ProductType": "Tablet"},
    {"ConsumerName": "Samuel", "Model": "MacBookAir",
     "ProductType": "Laptop"},
    {"ConsumerName": "Vikrant", "Model": "Edge60",
     "ProductType": "Smartphone"},
    {"ConsumerName": "Kiara", "Model": "Pad3",
     "ProductType": "Tablet"},
]
# Create DataFrame
Consumerdf = pd.DataFrame(ConsumerElectronics)
print("Consumer DataFrame:")
print(Consumerdf)
# Use of an "Is-a" Mapping
category_map = {
    "SimpleMobile": "Mobiles",
    "Smartphone": "Mobiles",
    "Laptop": "Portable Computing",
    "Tablet": "Portable Computing"
}
# New column created from generalisation of category
Consumerdf["New_Category"] =
Consumerdf["ProductType"].map(category_map)
print("DataFrame with New Category:")
print(Consumerdf)
```

Program 2: Use of Abstraction

In Program 2, *ConsumerElectronics* is a list of dictionary items containing data of 6 individuals. Using this data, a dataframe named *Consumerdf* has been created and printed (See Table 3). An “is a” hierarchy named *category_map* has been created, as per the example requirements. This mapping is then used to create a column named *New_Category* using the *ProductType* and *category_map*. The result of this operation is shown in Table 4.

Table 3: Consumer DataFrame

ConsumerName	Model	ProductType
Manish	Reno	Smartphone
Meenu	Thinkpad	Laptop
Rafiq	Galaxy	Tablet
Samuel	MacBookAir	Laptop
Vikrant	Edge60	Smartphone
Kiara	Pad3	Tablet

Table 4: DataFrame with New Category:

ConsumerName	Model	ProductType	New_Category
Manish	Reno	Smartphone	Mobiles
Meenu	Thinkpad	Laptop	Portable Computing
Rafiq	Galaxy	Tablet	Portable Computing
Samuel	MacBookAir	Laptop	Portable Computing
Vikrant	Edge60	Smartphone	Mobiles
Kiara	Pad3	Tablet	Portable Computing

Check your progress 2

1. What is the main purpose of using data semantics?

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2. Explain the "Is A" relationship (generalisation) with an example.

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3. How can you perform Is A mapping in data using Python?

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1.6 SET-BASED PROFILING

Set-based profiling is a data analysis technique that evaluates complete sets or tables of data rather than focusing only on individual rows or columns. By evaluating data against

standards helps in determining whether it is accurate, consistent, or contains null values. This method offers a more thorough and expansive comprehension of the dataset's general structure.

Set-based profiling finds linkages both within and between datasets as well as trends, correlations, and problems with data quality. It makes it possible to identify foreign key relationships and gain a deeper understanding of data dependencies, key candidates, and cross-table linkages. It is very helpful for large datasets, business intelligence, and data warehousing since it generates statistical summaries like frequency, mean, and minimum/maximum values.

The procedure begins with collecting data sources together with their accompanying information for analysis. It entails analysing the dataset to find dependencies and relationships, then cleaning the data to remove redundancy which guarantees a consistent structure. Data profiling technologies produce statistical measurements that aid in characterising and assessing the dataset once the data has been cleansed.

Here are a few straightforward instances of using set-based profiling to carry out tasks.

Compare the Overlap and Differences of Two Columns

The set intersection (&) and set difference (-) operators can be used to compare the set of values available in a data set in two different columns. The intersection and differences between the two sets made from the two dataframes are found in the example that follows.

Example: A university keeps individual student data every year. Determine the list of (i) students who are continuing from the previous year, (ii) students who have passed (assuming no dropouts), and (iii) new students who have joined using the data from the previous two years.

Solution: The first step is to generate data frames for both new and existing students. Making a set for every student in both the old and new batches is the next stage. The set intersection and difference operations are used to obtain the required student lists (refer to Program 3).

```
import pandas as pd
old_students = pd.DataFrame({'student_id': [101, 102, 103, 104]})
new_students = pd.DataFrame({'student_id': [103, 104, 105, 106]})
old_batch = set(old_students['student_id'])
new_batch = set(new_students['student_id'])

Continuing_student = old_batch&new_batch
print("Students who continued",Continuing_student)

Passed = old_batch - new_batch
print("Students who Passed",Passed)

New_student = new_batch - old_batch
print("Students who Joined",New_student)
```

Program 3: Set Intersection and Difference

The output of this program would be:

```
Students who continued {104, 103}
Students who Passed {101, 102}
Students who Joined {105, 106}
```

Foreign Key Validation (Referential Integrity)

The foreign key violations can also be determined by first creating the set of data and using the set difference (-) operator. The following example explains this.

Example: Students' data is electronically maintained by university. The data had an adequate backup. Nevertheless, a few fake records were added to the data. Use the correct data to verify the accuracy of this inaccurate data.

Solution: Program 4 does the same function as stated earlier. First, dataframes including both accurate and inaccurate student data are created. Next, a set is created for each of the student data; correct_set and incorrect_set. The list of students who were added wrongly is produced by using the set difference operator.

```
import pandas as pd
Incorrect_studentsdata = pd.DataFrame({'student_id': [101,
102, 103, 999]})
Correct_studentsdata = pd.DataFrame({'student_id': [101,
102, 103]})
invalid_ids = set(Incorrect_studentsdata ['student_id']) -
set(Correct_studentsdata ['student_id'])

print("Invalid foreign keys:", invalid_ids)
```

Program 4: Checking Referential Integrity

The output of this program is as follows:

```
Invalid foreign keys: {999}
```

In addition to set based profiling, you can perform the following types of data profiling.

Discovery based on Structure

This technique guarantees a data set's consistency in data format. There are several methods that can be used to analyse this type of profiling. Pattern matching is a crucial technique for format-specific data.

For instance, the structure discovery could identify which of the following information entered into a column name Email_Address is wrong.

Email_Address
mona@gmail.com
abc123
mary@yahoo.com

The data value "abc123" is an incorrect email address since it does not follow the intended email address pattern. Structure discovery will identify that the first two values follow the email format.

Content-driven Discovery

This kind of profiling involves looking for mistakes or systemic problems in data rows. This method can assist in identifying erroneous values by meticulously scrutinising each component of the data.

For example, the following column displays the students' grades for a certain subject:

200, 22, 25, 30, and NULL.

Two content incompatibilities are present here: first, the value 200 seems to be an outlier, and second, there is a NULL (missing data) value in the data. As a result, content mismatch can help identify outliers and missing values.

Relationship discovery

This entails determining the relationship between each group. To do this, analysts will first use metadata analysis to find the connections between data, after which they will concentrate on the correlations between specific areas. It involves determining functional dependencies, foreign key relationships, and primary key uniqueness.

For instance, there is a foreign key relationship between the Student and Account data sets.

Student	
Student id	Name
1	XX
2	YY

Account	
Account id	Student id
A101	1
A102	3

You can determine that there is an error in the Account data set because Student_id3 does not exist in the Student Dataset, thanks to the foreign key relationship.

1.7 DATA WRANGLING TOOLS

Another aspect of Data wrangling also concerns understanding the tools and technologies available to efficiently clean, structure, and carry out data enhancement. These tools exist in variety from basic spreadsheet programs to sophisticated platforms for data science. Some of the data wrangling tools are discussed in this section.

Recognising Your Needs

Before examining specific technologies, it is important to understand your data wrangling requirements. Consider volume and complexity of data, the kind of data sources you are working with, and whether you require automated processes or unique solutions.

Data Wrangling Tools

Data wrangling tools can be classified into the following broad categories:

- **Basic Tools:** These include spreadsheet programs that are easily usable and available for fundamental operations on data like sorting, filtering, and basic computations, such as Google Sheets and Microsoft Excel.
- **Programming:** The two most popular languages for data science are R and Python, offers extensive libraries and tools for data manipulation, including

dplyr and pandas, respectively. For complex data processing jobs, these languages offer accuracy, flexibility, and strength.

- **Dedicated Software for Data Wrangling:** Other programs created specifically for data wrangling include Alteryx, Talend, and Trifacta. They employ a graphical user interface (GUI) to smooth and improve the process of cleaning, transforming, and enriching data.

Platforms for Integrated Data Science: In addition to other data science features, platforms like as KNIME, RapidMiner, and Dataiku offer extensive environments with data wrangling skills. These solutions offer the entire range of accessible techniques and are appropriate for end-to-end data projects, from wrangling to modelling and visualisation.

The Future of Data Wrangling

Following future developments and trends that have an impact on data wrangling tools are:

1. **Machine learning and automation:** It is expected that the use of AI and machine learning technologies in data wrangling tools will increase, automating more complex data preparation procedures. This update aims to free up data scientists and analysts to work on more strategic projects by reducing time and effort needed for data transformation and cleansing.
2. **Increased Integration:** Companies are using a wide range of data sources, and the integration capacities of data wrangling systems will continue expanding. The integration will involve improved interface with databases, cloud storage, and other data formats, enabling a more efficient data pipeline from input sources.
3. **User-Friendly Interfaces:** To make data wrangling accessible to a wider audience, including those with no programming experience, we might anticipate tools with more user-friendly graphical user interfaces. These developments will democratise data analysis by empowering more organisational employees to participate in data-driven decision-making.
4. **Real-time Data Processing:** As the demand for real-time analytics increases, data wrangling technologies will enable processing of data in real-time. This feature can be very helpful for applications that need immediate insights, such as fraud detection, real-time consumer feedback analysis, and operational performance monitoring.
5. **Governance and Data Quality:** As organisations become more data-driven, data governance and quality will become increasingly important. To ensure that the data used for analysis is accurate, consistent, and compliant with legal standards, additional functionality will be added to data wrangling tools to automatically identify data quality issues and enforce data governance rules.
6. **Collaboration Features:** Improved data wrangling collaboration tools will enable teams to work simultaneously more effectively on data preparation activities in the future. Version control, shared workspaces, and the ability to track and examine changes made by different team members are a few of these capabilities.

7. **Integration of Advanced Analytics:** Technologies for data manipulation will gradually blend with advanced analytics capabilities such as machine learning algorithms and predictive modelling. This integration will simplify the process of turning raw data to useful insights, enabling users to move directly from data preparation to analysis.
8. **Pay attention to security and privacy:** As international data privacy rules and regulations become stricter, data wrangling solutions need to have strong security features and compliance processes. This leads to ensuring that tools follow legal requirements and that data is secured or anonymised as needed.

1.8 SUMMARY STATISTICS

A statistical summary gives information about the data in a sample. It may help with a more thorough understanding of the values. It may include a set of data's mean and standard deviation in addition to the total number of values, lowest value, and greatest value. This aids in understanding the distribution of values, patterns, and outliers within a data set. Better data analysis may result from this, which is especially useful when dealing with large amounts of data.

Summary statistics are essential for the preliminary investigation of data in data wrangling, which comprises the following:

- **Knowing Data:** They provide a synopsis of the data's central value, shape, and dispersion. For example, knowing a group's average (mean) age and age range can help you rapidly ascertain its composition.
- **Identifying Outliers:** Data points that deviate greatly from the norm are known as outliers. They can be found using summary statistics. For example, a dataset is most likely an outlier if its greatest value differs significantly from the mean or median. Outliers are often explicitly identified using the Interquartile Range (IQR).
- **Identifying Missing Values:** Summary statistics display the count of all the data points. If this count falls short of your expectations, it means that there are missing values that need to be fixed.
- **Assessing Data Distribution:** The distribution of the data can be inferred by comparing the mean, median, and mode.
 - The mean, median, and mode are roughly equal in a symmetrical (normal) distribution.
 - These numbers will be different in a skewed distribution. For instance, the mean will be higher than the median in a distribution that is skewed to the right.
- **Guiding Transformation:** Many statistical models assume that the data has a normal distribution. If summary statistics indicate that your data is excessively skewed, you can enhance model performance by applying a mathematical transformation (such as taking the log value) to make the distribution more symmetrical.

Summary statistics and data visualisation aid in transforming unprocessed numbers into insightful visuals:

- **Building Blocks for Visuals:** Most charts are visual summaries of statistics. A bar chart often displays the mean or a count. Proportions are shown using pie charts. In a box plot, the median, quartiles, and range are all shown concurrently.
- **Choice of Visuals:** You can choose the best chart type based on the initial summary statistics. A box plot is a good way to show the distribution and draw attention to outliers. If you want to examine the average values across numerous categories, a bar chart is a great choice.
- **Explaining Trends:** Summary statistics help in explaining the "why" in the underlying images. For example, a line graph of summary statistics can quantify a rising trend.
- **Communicating Insights:** Charts can communicate a lot of information. For example, instead of just showing increased sales data, a chart with summary statistics can communicate information like "Average monthly sales increased by 20% (from a mean of 10,000 to 12,000)". This is more powerful and precise.

You have already studied summary statistics in MCS-066.

Check your progress 3

1. Which type of data profiling focuses on identifying incorrect values in data rows?
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.....
2. Give an example of a chart that is a visual representation of summary statistics.
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.....
3. How do summary statistics help in communicating insights from a chart?
.....
.....
4. If the mean of a dataset is greater than its median, what can you infer about the data's distribution?
.....
.....
5. Which measure of central tendency is most affected by outliers? Why?
.....
.....

1.9 SUMMARY

Through error correction, data organisation, and missing value filling, data wrangling converts raw data into a format that can be used for analysis. This process enhances data quality, enabling more accurate insights and boosting the pipeline's overall effectiveness for data analytics. It is crucial due to the growing volume and variety of data, the need for superior input for speedier decision-making, compliance with data privacy laws, and

improved data quality and accuracy. Data wrangling includes the following essential steps: gathering, cleaning, organising, enhancing, verifying, storing, and recording. One of the many benefits of data wrangling is high-quality data, which is essential for quick and accurate decision making in the quickly growing globe.

Data semantics improves analysis, ensures consistency, facilitates integration, and clarifies data meaning by organising unstructured data using semantic models. Semantic models improve analysis, ensure consistency, facilitate integration, and clarify the meaning of data.

Set-based profiling identifies trends, correlations, and issues with data quality in addition to connections within and between datasets. It enables a deeper comprehension of data dependencies and cross-table linkages, as well as the identification of foreign key links. Large datasets, business intelligence, and data warehousing benefit greatly from its ability to produce statistical summaries such as frequency, mean, and minimum/maximum values.

A brief discussion of data wrangling tools is also included in this unit. For additional information on the newest data wrangling tools, you are encouraged to visit the WWW. For the initial analysis of data in data wrangling, summary statistics ideas are crucial.

1.10 ANSWERS AND SOLUTIONS

Check Your Progress 1

1. Why is Data Wrangling Important for AI and Machine Learning?

The accuracy of results of AI and machine learning (ML) models depends on the quality of data. Real-world data frequently contains errors, is duplicated, inconsistent, or incomplete. Thus, it is important to perform data wrangling so as to produce clean, precise, and well-organized data. In other words, data wrangling prepares data prior to model training.

Importance:

- Improves data quality and accuracy
- Removes missing or incorrect values
- Converts raw data into structured format
- Increases model performance and prediction accuracy
- Saves time during analysis and model building

Without proper data wrangling, Machine Learning models may give poor or misleading results.

2. What is the Primary Goal of Data Wrangling? Define Data Profiling.

The primary goal of data wrangling is to transform unstructured, raw data into an understandable format that can be utilized for reporting, analysis, or machine learning.

Data Profiling is Examining and analyzing data to comprehend its structure, quality, patterns, and linkages is known as data profiling. Data profiling guarantees that the dataset is appropriate for additional processing and analysis by identifying missing values, duplicate records, data kinds, inconsistencies, errors, and anomalies.

3. Why is Data Wrangling Needed?

Data wrangling is required as raw data gathered from many sources is typically contains errors and is inappropriate for direct use, Data wrangling helps businesses get trustworthy insights from their data by cleaning up erroneous or missing data, eliminating duplicate records, combining data from various sources, converting data into a standard format, improving data consistency and quality, preparing data for analysis and machine learning, and increasing efficiency and accuracy in decision-making.

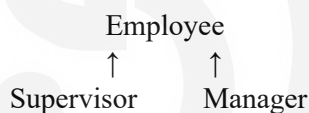
Check Your Progress 2

1. What is the Main Purpose of Using Data Semantics?

Primary objective of data Semantics is to give data a clear meaning and context, which makes it easier for people and computer systems to comprehend, distribute, and analyse data. Data semantics enhances data comprehension, facilitates the integration of data from many sources, reduces ambiguity and uncertainty, ensures consistent data interpretation, improves system communication, and increases analysis accuracy.

2. Explain the "Is A" Relationship (Generalisation) with an Example.

A generalisation relationship in which one thing is a specialised form of another is represented by the "Is A" relationship. It displays how objects are classified or inherited. For example, in the statement - “A *supervisor* is an *employee*, and a *manager* is also an *employee*.”, *employee* is a general class and *supervisor* and *manager* are specialised classes. It can be represented with the help of following diagram:



This means a supervisor or a manager inherits the properties of class Employee

3. How can you perform Is A mapping in data using Python?

You can create an is-a mapping from specialised classes to general classes, as shown in Program 2, which can then be used to categorise data on the basis of generalised classes.

Check Your Progress 3

1 Which type of data profiling focuses on identifying incorrect values in data rows?

Finding erroneous values in data rows is the main goal of content profiling. Additionally, it aids in the detection of wrong data, inconsistent entries, duplicate data, improper formats, and missing values. For instance, suppose a dataset's height in cm column has the following values:

166, 175, ABC, -6 content profiling recognises "ABC" and "-6" as erroneous and invalid values.

2. Give an example of a chart that is a visual representation of summary statistics.

Box Plot (Box-and-Whisker Plot) is a graphic representation of summary data. A box plot shows a summary of the data, including the median, outliers, Q1 (first quartile), Q3 (third quartile), and lowest and maximum values.

3. How do summary statistics help in communicating insights from a chart?

The five point summary (Minimum, Q1, Median, Q3 and maximum) and mean condense vast volumes of data into significant values and standard deviation is used to measure distribution of data. Some of the benefits of these statistics are:

- You can determine patterns and trends
- You can compare several datasets
- You can find probable outlier values in a data set
- You can recognise the distribution of data without examining each individual data point.

4. If the mean of a dataset is greater than its median, what can you infer about the data distribution?

The dataset is typically positively skewed or right skewed if the mean is higher than the median, which indicates that a few extremely large values are pushing the mean to the right. Most data values are concentrated on the lower end.

5. Which measure of central tendency is most affected by outliers? Why?

Since the mean is computed utilising every value in the dataset, it is the central tendency metric that is most impacted by outliers. Outliers, or extremely high or low values, can greatly raise or lower the average.

1.11 REFERENCES

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